

Solutions to Jan 26

① $(x^2+x+1)(x^2+x) = -2x-2$, $(x-1)^{-1} = -\frac{x^2}{5} - \frac{2x}{5} - \frac{3}{5}$

② $F \subset F(x^2) \subset F(x)$; $[F(x):F] = [F(x):F(x^2)][F(x^2):F]$
 $[F(x):F(x^2)] = 1$ or 2 , but $[F(x):F]$ is odd and so
 $[F(x):F(x^2)]$ divides an odd number so it is 1 so $F(x) = F(x^2)$

③ Let $h = \text{Irr}(f, F(x), X)$. We know $h \mid g$, so it is enough to show $\deg h = \deg g$ (hence $h = g$ is irr.)
 Now $[F(x, B):F] = [F(x, B):F(x)] [F(x):F] = [F(x, B):F(x)] [F(B):F]$
 so $[F(x):F] = \deg f$ & $[F(B):F] = \deg g$ divide $[F(x, B):F]$
 but these numbers are relatively prime so $\deg f \mid \deg g$
 divides $[f(x, B):F] = [F(x, B):F(x)] [F(x):F] = \deg h \deg f \leq \deg g \deg f$
 so $\deg h = \deg g$ $\square$

④ $\alpha = \sqrt[4]{2}$ has $\text{irr}(\alpha, \mathbb{Q}, x) = x^4 - 2$ (irr by Eisenstein)
 Now if $\mathbb{Q}(\alpha) \supsetneq F \supsetneq \mathbb{Q}$ then $[F:\mathbb{Q}]$ divides $[\mathbb{Q}(\alpha):\mathbb{Q}] = 4$
 so $[F:\mathbb{Q}] = 2$ so $F = \mathbb{Q}(\sqrt{d})$ some $d \in \mathbb{Q}$ (easy to check)

$\text{irr}(\alpha, F, x)$ has degree 2, divides $x^4 - 2$. Say $\text{irr}(\alpha, F, x) = x^2 + ax + b$. Then $F \supset \mathbb{Q}(a, b)$ and $\mathbb{Q}(a, b) \not\subset \mathbb{Q}$ so
 $F = \mathbb{Q}(a, b)$. $(x^4 - 2) = (x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - i\sqrt[4]{2})(x + i\sqrt[4]{2})$
 $(x + i\sqrt[4]{2})(x - i\sqrt[4]{2}) = x^2 - \sqrt{2}$ gives $F = \mathbb{Q}(\sqrt{2})$
 $(x + i\sqrt[4]{2})(x - i\sqrt[4]{2}) = x^2 + \sqrt{2}$ gives $F = \mathbb{Q}(\sqrt{2})$
 $(x \pm i\sqrt[4]{2})(x \pm i\sqrt[4]{2}) = x^2 + (\pm i\sqrt[4]{2} \pm i\sqrt[4]{2})x \pm \sqrt{2} \cdot i$ gives $\mathbb{Q}(i\sqrt{2}, \pm i\sqrt[4]{2} \pm i\sqrt[4]{2}) \not\subset \mathbb{Q}(a, b)$
 which is not allowed.

⑤ $G: \mathbb{Q}(x) \rightarrow \mathbb{C}$ fixes $\mathbb{Q}$
 and is determined by $G(\alpha) = \alpha, \alpha^2, \alpha^4, \alpha^5, \alpha^7, \alpha^8$ (the other
 roots of $x^6 + x^3 + 1$ i.e. the other primitive 9th roots of 1)